

Explainable AI for Multi-Class Obesity Prediction Using Machine Learning Models

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Abstract—Obesity has become a world wide public health issue presenting trends of chronic diseases and lifestyle consequences. The ability to predict and categorize obesity grades accurately would help in the early intervention and also in the provision of individual approach to therapy. The given research provides a multi-class system of obesity classification grounded on the UCI Obesity Dataset, including 2,111 records with 17 demographic, nutritional, and lifestyle factors. The data was preprocessed, feature scaled and class imbalance was alleviated using SMOTE and a class-weighted loss. Six machine-learning models, such as logistic regression, random forest, XGBoost, LightGBM, CatBoost, and support vector machine, were trained and hyperparameter optimization was done via GridSearchCV. Accuracy, precision, recall, F1 -score, confusion matrices and ROC-AUC were used to assess model performance. The maximum accuracy of 96.22% was obtained by LightGBM and SVM. Explainable artificial-intelligence (especially SHAP), was used to analyze model results, and the weight, nutrition, and physical activity were found to be the leading factors of obesity classification. The future work will concentrate on federated learning and real-time implementation to help facilitate the process of scalable privacy-preserving obesity prediction.

Index Terms—Obesity, Machine learning, Explainable AI, Multi-class obesity prediction, LightGBM, Support Vector Machine, UCI Obesity Dataset.

I. INTRODUCTION

Obesity has become one of the foremost global public health issues of the 21st century, with a fast rising prevalence in both industrialized and developing nations [1]. Obesity, defined by an excessive buildup of body fat, is a critical medical condition that markedly increases the risk of chronic diseases, including type 2 diabetes, cardiovascular problems, hypertension, sleep apnea, and some malignancies [2], [3]. The World Health Organization indicates that global obesity rates have nearly tripled since 1975, impacting millions across all age demographics [4], [5]. The etiology of obesity is multifaceted, encompassing intricate interactions among genetic, behavioral, environmental, and lifestyle determinants [6]. Specifically, detrimental eating habits, lack of physical activity, sedentary lifestyles, and heightened intake of energy-

dense meals exert a significant and alterable influence [7]. Conventional analytical methods frequently fail to elucidate these intricate interactions [8]. Recently, machine learning methodologies have demonstrated significant potential in the analysis of extensive health data, facilitating precise obesity prediction and risk evaluation [9], [10]. This study employs and assesses various machine learning models to forecast obesity levels based on lifestyle and nutritional characteristics, with the objective of enhancing early detection and preventive decision-making [11].

II. LITERATURE REVIEW

Recent studies have shown the efficacy of machine learning methods for classifying obesity based on food, lifestyle, and demographic information [12], [13]. Asiff et al. (2025) utilized Decision Tree, SVM, Random Forest, and Extra Tree classifiers on a dietary and demographic dataset, with the Extra Tree model attaining the highest accuracy of 95% [4]. Shaban et al. (2025) introduced the ObeRisk framework, which integrates EC-QBA-based feature selection with ensemble learning utilizing several machine learning models, attaining an exceptional accuracy of 96%, surpassing the majority of current methodologies [5]. Gutiérrez-Gallego et al. (2024) examined sociodemographic and lifestyle data from Madrid inhabitants employing traditional machine learning methodologies, wherein a cascade classifier integrating Gradient Boosting, Random Forest, and Logistic Regression achieved an accuracy of 79% [6]. Khater et al. (2023) concentrated on lifestyle and dietary practices, excluding BMI characteristics, and achieved 75% accuracy using an interpretable machine learning model, highlighting the impact of behavior [7]. Yagin et al. (2023) utilized neural networks with refined feature selection and Bayesian optimization, attaining an accuracy of 93.06% [8]. Numerous research emphasized the resilience of Random Forest models. Dirik et al. (2023) [9] demonstrated 95.78% accuracy utilizing Random Forest on physical and dietary datasets, whereas Kitis et al. (2023) [14] and Huang et al. (2022) [15] attained 95.78% and 95.60% accuracy, respectively, on the UCI Obesity Dataset.

III. METHODOLOGY

This section delineates the process for multi-class obesity categorization employing machine learning techniques. The process encompasses dataset preprocessing, exploratory analysis, model training, evaluation, and the use of explainable AI methodologies to guarantee dependable and reproducible outcomes.

A. Exploring the Dataset

This research used the Obesity Dataset from the UCI Machine Learning Repository [1], including 2,111 instances and 17 characteristics gathered from people in Mexico, Peru, and Colombia. The goal variable NOBeyesdad delineates seven categories of obesity. Of the data acquired, 23% originated from actual users, while 77% was artificially created using Weka with the SMOTE filter.

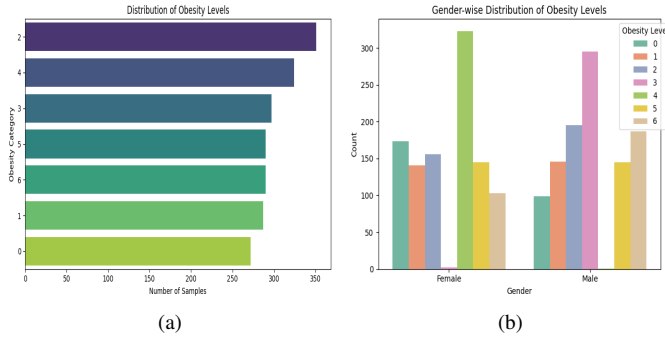


Fig. 1: (a) Distribution of Obesity Levels, (b) Gender-wise Distribution of Obesity Levels

B. Data Preprocessing

Data preprocessing was conducted to guarantee model robustness and consistency. Categorical variables were converted by label encoding and one-hot encoding, and numerical properties were normalized using StandardScaler. Outliers were examined using statistical methodologies. The dataset was then divided between 80% training and 20% testing sets, maintaining class distribution.

C. Exploratory Data Analysis (EDA)

An exploratory data analysis was performed to comprehend feature distributions and class connections. Class distribution plots, boxplots, correlation heatmaps, and scatter plots were used to examine demographic, lifestyle, and physiological variables. The investigations yielded insights into feature significance and informed further model selection and optimization.

D. Model Descriptions

This research utilizes various supervised machine learning methods to tackle the multi-class obesity classification issue.

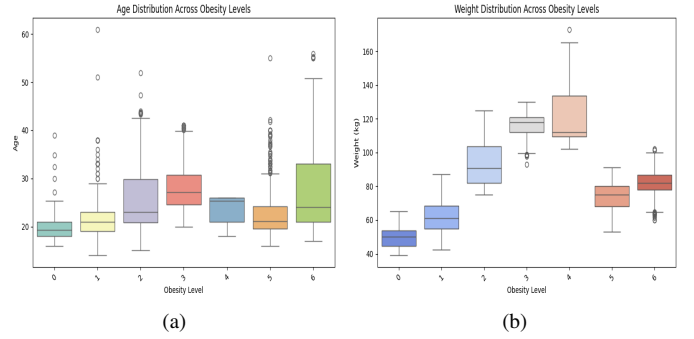


Fig. 2: (a) Age Distribution Across Obesity Levels, (b) Weight Distribution Across Obesity Levels

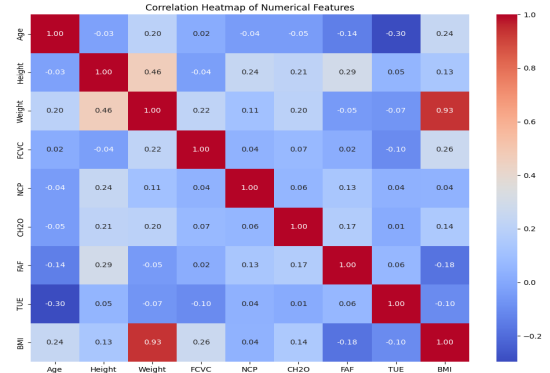


Fig. 3: Correlation Heatmap of Numerical Features

Logistic Regression (Softmax): Multinomial logistic regression utilizes the softmax function to model the posterior probability of each obesity class.:

$$P(y = k|x) = \frac{e^{\beta_k^T x}}{\sum_{j=1}^K e^{\beta_j^T x}} \quad (1)$$

It functions as a linear reference for multi-class classification.

Support Vector Machine (One-vs-Rest): SVM formulates ideal hyperplanes by maximizing the separation between

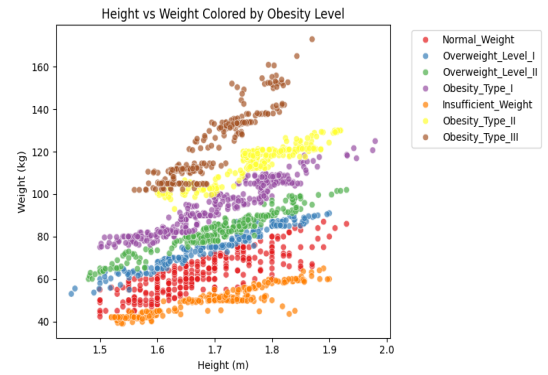


Fig. 4: Height vs Weight Colored by Obesity Level

classes:

$$\min \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \quad (2)$$

The OvR approach disaggregates the multi-class task into binary classifiers.

Random Forest: Random Forest is an ensemble method including decision trees that are trained by bootstrap sampling and feature randomness:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_n(x)\} \quad (3)$$

It enhances generalization and mitigates overfitting.

XGBoost: XGBoost constructs trees in a sequential manner by optimizing a regularized objective function:

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \quad (4)$$

It effectively captures intricate nonlinear interactions.

LightGBM: LightGBM employs histogram-based learning and leaf-wise tree development to enhance speed and precision:

$$\text{Gain} = \frac{G_L^2}{H_L} + \frac{G_R^2}{H_R} - \frac{(G_L + G_R)^2}{H_L + H_R} \quad (5)$$

It is ideally suited for extensive structured data.

CatBoost: It manages categorical features with ordered target statistics and symmetric trees:

$$\hat{y} = \sum_{k=1}^N f_k(x) \quad (6)$$

It mitigates prediction bias and enhances performance on heterogeneous datasets.

E. Multi-class Machine Learning Models with GridSearchCV

Numerous multi-class machine learning models were built utilizing GridSearchCV for hyperparameter optimization. Logistic Regression, Support Vector Machine (One-vs-Rest), Random Forest, XGBoost, LightGBM, and CatBoost were assessed by cross-validation. This technique guarantees equitable comparison and improves model generalization for obesity level classification.

F. Class Imbalance Handling

Data-level and algorithm-level strategies were employed to rectify class imbalance in the obesity dataset. The SMOTE technique was utilized to create synthetic samples for minority classes, while class-weighted learning was applied in specific models. The performance of the model was assessed prior to and subsequent to balancing to determine its effect on classification accuracy.

G. Evaluation Metrics

To thoroughly assess the efficacy of multi-class obesity categorization, various standard indicators were utilized.

Accuracy: Accuracy quantifies the ratio of correctly identified samples to the total number of cases:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

Precision (Macro & Weighted): Precision assesses the accuracy of positive predictions, averaged across different classes:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

Recall (Macro & Weighted): Recall assesses the model's capacity to accurately identify true positive events:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

F1-Score (Macro & Weighted): The F1-score signifies the harmonic mean of precision and recall, effectively balancing both measurements:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (10)$$

Confusion Matrix: The confusion matrix offers a detailed overview of accurate and inaccurate predictions for each class:

$$CM(i, j) = \text{Number of instances of class } i \text{ predicted as class } j \quad (11)$$

ROC-AUC (One-vs-Rest): ROC-AUC assesses the balance between true positive and false positive rates for each class:

$$\text{TPR} = \frac{TP}{TP + FN} \quad (12)$$

$$\text{FPR} = \frac{FP}{FP + TN} \quad (13)$$

H. Explainable AI

SHAP (SHapley Additive Explanations) was utilized to interpret model predictions, providing both global and local explanations. The significance of features was assessed to evaluate the impact of nutrition, physical activity, and weight on obesity classification. SHAP values offer quantitative insights into the contribution of individual attributes to each class prediction, hence augmenting transparency and trust in model judgments.

I. Overall Proposed Methodology

The proposed methodology encompasses data collection, preprocessing, exploratory analysis, management of class imbalance, multi-class model training with hyperparameter optimization, and evaluation through accuracy, F1-score, confusion matrix, and ROC-AUC. Explainable AI methodologies, such as SHAP and feature importance analysis, are utilized to elucidate model predictions. This approach guarantees a robust, reproducible, and interpretable classification of obesity.

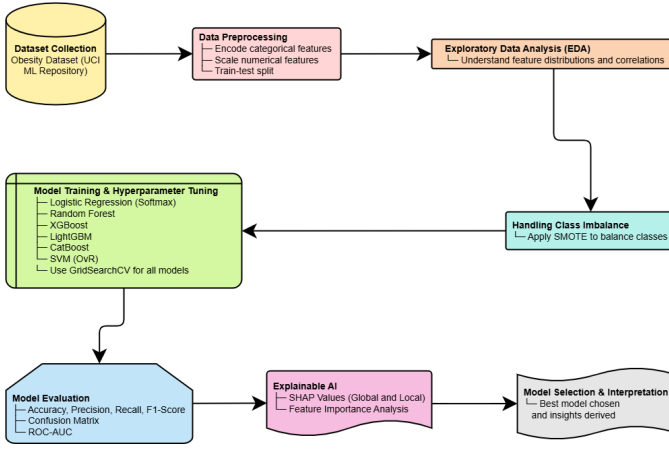


Fig. 5: Overall Proposed Methodology Diagram

IV. RESULTS AND DISCUSSIONS

This section illustrates the results of multi-class obesity classification utilizing the proposed machine learning models. It delineates the performance of each model, encompassing classification metrics, confusion matrices, ROC-AUC analysis, and insights from explainable AI. Observations and interpretations are presented to elucidate model behavior and the impact of features on obesity prediction.

A. Experimental Environment Setting

All of the experiments were performed on a laptop including an Intel Core i5-4210U processor at 2.40 GHz, 6 GB of RAM, and a 238 GB SSD, operating on a 64-bit OS. Model implementation and assessment were conducted utilizing Python 3 and conventional machine learning libraries.

B. Result Analysis of Obesity Classifications

The multi-class obesity classification models attained elevated accuracy across all categories. Logistic Regression obtained 92.67%, Random Forest 94.56%, XGBoost 95.27%, CatBoost 95.74%, and both LightGBM and SVM reached 96.22%. Each model successfully recognized individuals across seven obesity categories, showcasing their capacity to discern intricate correlations among demographic, lifestyle, and physiological characteristics for precise classification.

C. Classification Report

The classification report indicates precision, recall, and F1-score for each obesity category across all models. Metrics demonstrate consistently good performance, with the majority of models attaining weighted and macro averages exceeding 0.93, signifying precise identification of all seven obesity categories.

D. Confusion Matrix

The confusion matrices illustrate accurate and inaccurate predictions among obesity categories. LightGBM and SVM demonstrate significant diagonal dominance, whereas Logistic Regression and Random Forest display slight confusion between Normal Weight and Overweight_Level_I.

TABLE I: Specific Accuracy and Best Hyperparameters of ML Models

Model	Best Hyperparameters	Accuracy (%)
LightGBM	learning_rate=0.1, n_estimators=200, num_leaves=50	96.22%
SVM	C=10, kernel=linear	96.22%
CatBoost	depth=6, iterations=300, learning_rate=0.1	95.74%
XGBoost	learning_rate=0.05, max_depth=6, n_estimators=200, subsample=0.8	95.27%
Random Forest	max_depth=None, min_samples_split=2, n_estimators=200	94.56%
Logistic Regression	C=10	92.67%

TABLE II: Classification Metrics per Obesity Class for All Models (Class Labels: 0 = Insufficient Weight, 1 = Normal Weight, 2 = Obesity Type I, 3 = Obesity Type II, 4 = Obesity Type III, 5 = Overweight Level I, 6 = Overweight Level II)

Model	Class	Precision	Recall	F1-Score	Support
XGBoost	0	0.98	0.93	0.95	54
	1	0.84	0.97	0.90	58
	2	0.97	0.96	0.96	70
	3	0.98	0.98	0.98	60
	4	1.00	0.98	0.99	65
	5	0.93	0.88	0.90	58
	6	0.98	0.97	0.97	58
LightGBM	0	0.96	0.93	0.94	54
	1	0.86	0.97	0.91	58
	2	0.99	0.96	0.97	70
	3	0.98	0.98	0.98	60
	4	1.00	0.98	0.99	65
	5	0.98	0.91	0.95	58
	6	0.97	1.00	0.98	58
CatBoost	0	1.00	0.93	0.96	54
	1	0.85	0.97	0.90	58
	2	0.99	0.97	0.98	70
	3	0.98	0.98	0.98	60
	4	1.00	0.98	0.99	65
	5	0.93	0.88	0.90	58
	6	0.97	0.98	0.97	58
SVM	0	0.96	1.00	0.98	54
	1	0.96	0.91	0.94	58
	2	0.97	0.96	0.96	70
	3	0.97	1.00	0.98	60
	4	1.00	0.98	0.99	65
	5	0.92	0.95	0.93	58
	6	0.95	0.93	0.94	58

E. Receiver Operating Characteristic (ROC)–AUC Analysis

The ROC–AUC curves demonstrate significant class separability for all assessed models. LightGBM and SVM attained the maximum discrimination, with an estimated AUC of roughly 0.99, while CatBoost and XGBoost followed closely at around 0.98. The Random Forest model achieved an AUC of approximately 0.96, but the Logistic Regression model exhibited a poorer performance with an AUC of around 0.93.

F. Feature Importance and Explainability Using SHAP (XAI)

SHAP was utilized on the XGBoost classifier to improve model transparency by providing global and local explanations. The SHAP interaction analysis indicates that Weight,

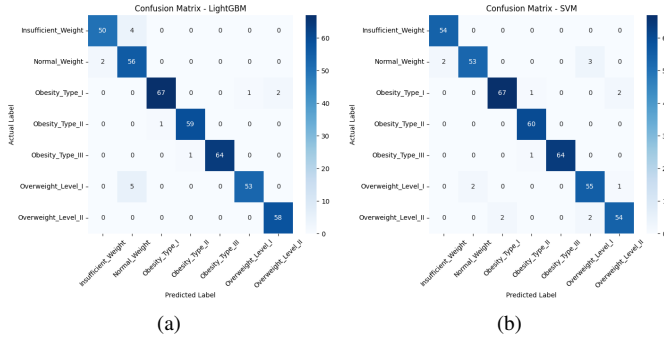


Fig. 6: Confusion Matrix of (a) LightGBM and (b) SVM

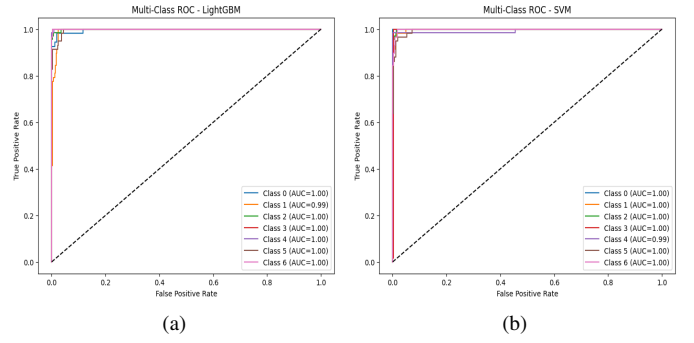


Fig. 9: ROC-AUC of (a) LightGBM and (b) SVM

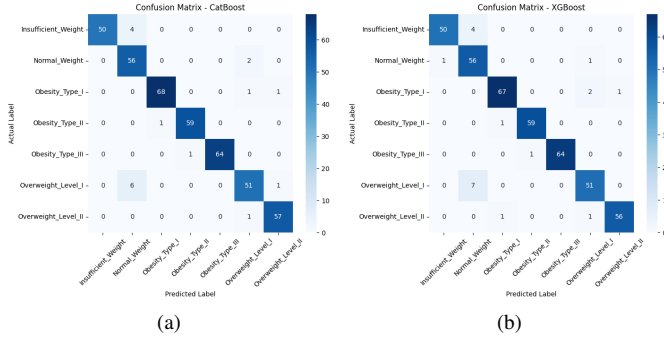


Fig. 7: Confusion Matrix of (a) CatBoost and (b) XGBoost

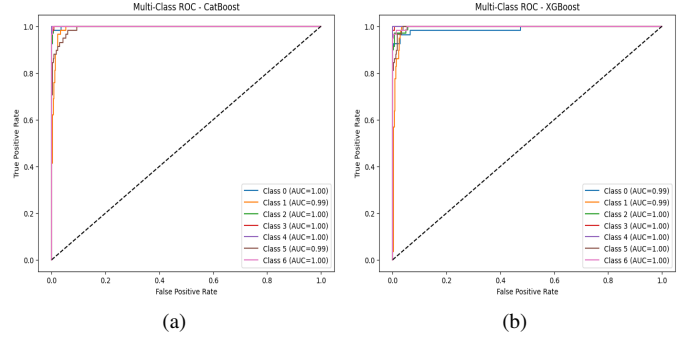


Fig. 10: ROC-AUC of (a) CatBoost and (b) XGBoost

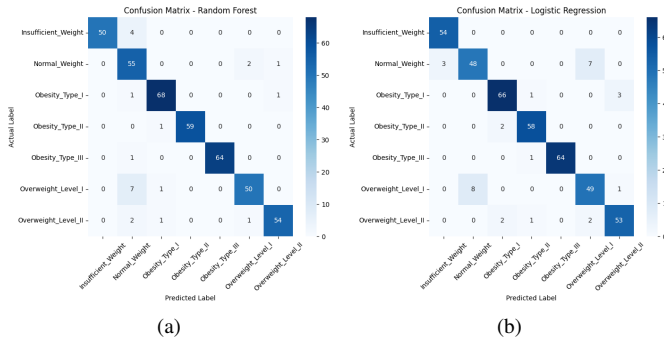


Fig. 8: Confusion Matrix of (a) RF and (b) LR

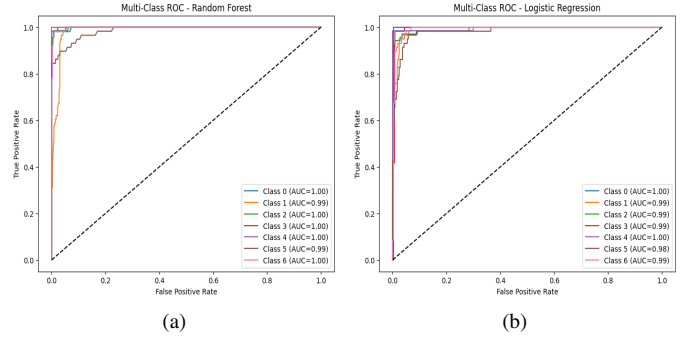


Fig. 11: ROC-AUC of RF and (b) LR

Age, and Height collectively have significant positive impacts on obesity-related classifications, whereas Physical Activity Frequency (FAF) exhibits a protective effect by diminishing risk in several instances. Water intake (CH2O) exhibits stable, low-variance interactions, signifying a consistent effect across samples. The feature importance plot underscores Gender, FCVC (vegetable consumption), Weight, and CAEC (eating between meals) as primary predictors. These elucidations affirm that the model's determinations correspond with established lifestyle and physiological aspects associated with obesity.

G. Discussion and Comparison

The experimental results indicate that LightGBM and SVM attained the maximum performance, each achieving an accuracy of 96.22%, signifying robust generalization across all obesity categories. Logistic Regression exhibited the lowest accuracy (92.67%), primarily attributable to its linear assumption and constrained capacity to delineate intricate class borders. In general, ensemble and boosting-based models regularly surpassed conventional methods. In comparison to recent studies, the proposed framework outperforms the majority of existing methods and marginally enhances the ObeRisk model, validating the efficacy of integrating sophisticated machine

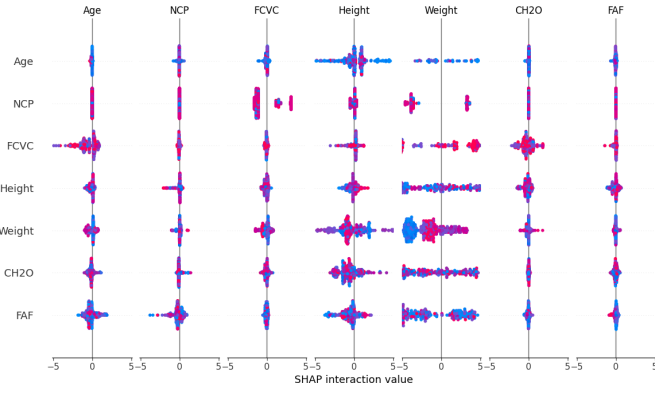


Fig. 12: SHAP Interaction Value Matrix

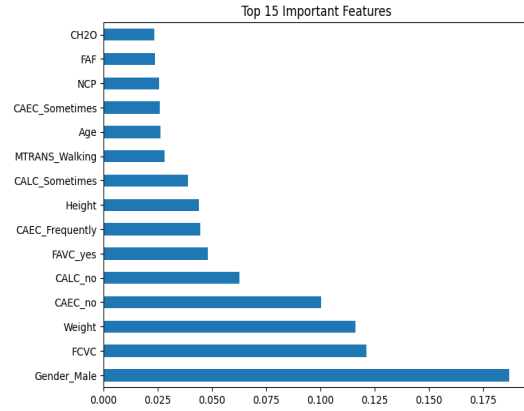


Fig. 13: Global Feature Importance Plot

learning techniques for obesity categorization.

V. CONCLUSION

This study introduced a thorough multi-class obesity categorization system employing both traditional and sophisticated machine learning algorithms. Experimental findings indicate that gradient boosting-based methods and SVM exhibit robust and consistent performance, with a maximum accuracy of 96.22%, whereas linear models have relatively diminished efficacy due to intricate feature interactions. The incorporation of explainable AI (SHAP) yielded significant global and local insights into essential lifestyle and physiological determinants affecting obesity. This framework will be augmented with Federated Learning and implemented in real-time health monitoring apps in future endeavors.

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TABLE III: State-of-the-Art Comparison on Obesity Classification

Ref.	Dataset	Methods Applied	Best Model (Accuracy)
[4]	UCI Obesity Dataset	LR, LGBM, XGBoost, AdaBoost, MLP, KNN, SVM, ObeRisk (EC-QBA)	ObeRisk (96%)
[6]	Lifestyle & Health Dataset (Madrid Residents)	Neural Network, Chi-square, F-Classify, Mutual Information	Neural Network (93.06%)
[9]	Physical & Dietary Habits Dataset	MLP, SVM, FuzzyNN, FURIA, RS, RT, RF, NB, LR, DT	Random Forest (95.78%)
Proposed Work	UCI Obesity Dataset	LR, RF, XGBoost, LGBM, CatBoost, SVM	LGBM & SVM (96.22%)

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